CORAL run 2026-10-11_130334 — results (all seeds)

Run summary

Item	Value
Task	cache-ext-get-scan (LevelDB GET/SCAN, cache_ext eviction-policy search)
Setup	Branch absolute-throughput-score: all seven reference policies (fifo, mru, s3fifo, sampling, get_scan, mglru, lhd) are available to the agent; the starting file is the noop policy. Score = measured throughput in ops/s (absolute; higher is better), no baseline calibration.
Run start → stop (local)	13:03:34 → 13:32:34 (29.0 min)
Stop reason	max_real_attempts (cap 20, real attempts 20); triggering attempt 7f1ae77b, score 1246.43
Agent	1 agent (captain-nemo), runtime claude_code, model alias claude-opus-5-5 → Bedrock us.anthropic.claude-opus-5-5
Attempts recorded	20 = 20 real + 0 --tune (tune attempts are graded identically but do not count toward the cap); 1 of them crashed without a score (#10); coral eval invocations: 24
Attempts by code	7 ran an unmodified reference policy (#1 get_scan.c, #2 s3fifo.c, #9 mglru.c, #12 mglru.c, #13 fifo.c, #14 sampling.c, #15 lhd.c); 12 scored agent-evolved variants
Attempt status (CORAL)	improved: 5, regressed: 14, crashed: 1 (relative to the agent's own previous best)
Agent sessions (claude -p)	22; terminal reason: aborted_streaming 16, aborted_tools 4, completed 2
LLM usage (agent)	29,750 output tokens; 5,309,724 cache-read + 176,266 cache-write + 140 input tokens; Claude Code cost estimate $2.54; API time 656 s
Notes / skills written	0 notes saved (4 agent actions were denied by the permission layer); skills present: create-notes, deep-research, organize-files, skill-creator

Headline numbers

Metric	Value	Metric	Value
Unmodified S3-FIFO, this run	1,216.66 ops/s	Unmodified S3-FIFO, all runs pooled (n=6)	1,241.4 ± 20.2 ops/s (range 1,216.7 … 1,263.0)
Highest score (any attempt)	#9 3233ed4a mglru.c (unmodified): 1,262.31 ops/s (+1.68% vs pooled S3-FIFO)	Best agent-evolved attempt	#20 7f1ae77b: 1,246.43 ops/s (+0.40% vs pooled S3-FIFO)
Unmodified references in this run	get_scan 1,067, s3fifo 1,217, mglru 1,262, mglru 1,213, fifo 1,227, sampling 1,115, lhd 1,103	Evolved attempts above pooled S3-FIFO mean	2 / 12 (above its pooled max: 0)
Throughput, mean ± sd of 19 scored	1,203.6 ± 53.5 ops/s	Worst score	#1 get_scan.c (unmodified): 1,066.96 ops/s
Final attempt (#20)	evolved: 1,246.43 ops/s	Reference: earlier noop baseline (run 1, external)	1,224.3 ± 9.5 ops/s (n=8)

The reference for every Δ in this report is the mean of all unmodified S3-FIFO measurements (this run plus the three earlier runs), a common yardstick across experiments. Single evals of identical code vary by roughly 1–4% here, so a Δ of that size is not evidence of improvement; the noise section quantifies it.

Scoring and recorded probes

Probe	Source	Role in this run	Unit
throughput	results.json: throughput_ops_per_sec	THE SCORE (maximized)	ops/s
rbytes	cgroup io.stat (delta over measured phase)	diagnostic only	MiB
wbytes, rios	cgroup io.stat	diagnostic only	MiB, count
refault_file, pgmajfault, pgactivate, resident_file	cgroup memory.stat	diagnostic only	count, count, count, MiB
wallclock	benchmark wall time (setup + warm-up + measured)	diagnostic only	s
policy_counters	metrics file a policy may export (none did in this run)	diagnostic only	count

Environment and workload

Evaluation node (CloudLab, Wisconsin)

Component	Value	Component	Value
Node type	c220g1	Kernel	6.6.8-cache-ext+ (Ubuntu 22.04.2)
CPU	2 × Intel Xeon E5-2630 v3 @ 2.40 GHz	Cores / threads	16 cores / 32 threads (2 sockets, 2 NUMA nodes)
Memory	131,732,952 kB (125.6 GiB), no swap	Disks	2 × 1.1 TB SAS HDD, 1 × 447 GB SSD
Benchmark filesystem	/mydata on /dev/sda3 (HDD), 63 GB	Cgroup	cgroup v2; benchmark memory.max = 64 MB
Compiler / tools	clang 14.0.0-1ubuntu1.1, bpftool v7.3.0	Eval isolation	one eval at a time (flock); policy attached via BPF struct_ops

Workload parameters (benchmark harness defaults, every attempt)

Parameter	Value	Parameter	Value
Benchmark	My-YCSB (leveldb-scan branch), run_leveldb	Request distribution / Zipfian constant	zipfian / 0.99
Operation mix	95% GET (read), 5% SCAN	Worker threads	4 (exactly 1 dedicated scan thread)
Scan length	500 keys per SCAN	Measured phase	30 s
Database entries	500,000	Warm-up phase	10 s (same mix, probes not recording)
Key / value size	16 B / 200 B	LevelDB block cache	16 MiB
Cgroup memory limit (page cache)	64 MB	Page cache at start	dropped before each run
Per-eval timeout	180 s	Observed eval wall time	39.7 s mean (39.5 … 40.0)

Per-attempt results

#	Local	Commit	Parent	Policy	Class / status	SCORE = throughput (ops/s)	Δ ops/s vs S3-FIFO mean	Δ%	Wall s	Eval s	+/− lines	Code lines
1	13:04:35	9d8172f7	5cefdb5c	get_scan.c (unmodified)	real / improved	1,066.96	-174.5	-14.05	39.8	45	+509/−50	603
2	13:06:35	5f742d7a	9d8172f7	s3fifo.c (unmodified)	real / improved	1,216.66	-24.8	-1.99	39.7	46	+409/−410	602
3	13:08:28	463657d3	5f742d7a	evolved	real / regressed	1,177.20	-64.2	-5.17	39.9	46	+83/−3	682
4	13:09:40	63f83a6a	463657d3	evolved	real / improved	1,234.54	-6.9	-0.55	39.8	46	+1/−1	682
5	13:10:50	f805b52d	63f83a6a	evolved	real / regressed	1,210.02	-31.4	-2.53	39.8	46	+1/−1	682
6	13:11:51	2b1dc4db	f805b52d	evolved	real / improved	1,240.09	-1.3	-0.11	39.9	46	+1/−1	682
7	13:13:06	545a9bfa	2b1dc4db	evolved	real / regressed	1,232.56	-8.9	-0.71	39.8	46	+15/−0	697
8	13:14:34	2403005d	545a9bfa	evolved	real / regressed	1,232.56	-8.9	-0.71	39.9	46	+1/−1	697
9	13:16:11	3233ed4a	2403005d	mglru.c (unmodified)	real / improved	1,262.31	+20.9	+1.68	39.6	47	+863/−451	1109
11	13:18:38	61e5e18b	dbea77c9	evolved	real / regressed	1,233.15	-8.3	-0.67	39.6	46	+4/−4	1188
12	13:19:38	19e7b6b1	61e5e18b	mglru.c (unmodified)	real / regressed	1,212.78	-28.6	-2.31	39.7	46	+5/−84	1109
13	13:20:54	44a8a8a9	19e7b6b1	fifo.c (unmodified)	real / regressed	1,227.31	-14.1	-1.14	39.6	46	+103/−941	271
14	13:23:51	11bac9cf	44a8a8a9	sampling.c (unmodified)	real / regressed	1,114.84	-126.6	-10.20	39.7	46	+300/−107	464
15	13:25:49	7efd90dc	11bac9cf	lhd.c (unmodified)	real / regressed	1,103.11	-138.3	-11.14	39.5	46	+591/−269	786
16	13:27:15	d799dfdb	7efd90dc	evolved	real / regressed	1,242.21	+0.8	+0.06	39.6	46	+382/−468	700
17	13:28:24	ca44f70f	d799dfdb	evolved	real / regressed	1,205.25	-36.2	-2.91	39.6	46	+1/−1	700
18	13:29:25	9279fd33	ca44f70f	evolved	real / regressed	1,234.39	-7.0	-0.57	39.9	46	+1/−1	700
19	13:30:39	d7fd5641	9279fd33	evolved	real / regressed	1,176.66	-64.8	-5.22	40.0	46	+1/−1	700
20	13:31:43	7f1ae77b	d7fd5641	evolved	real / regressed	1,246.43	+5.0	+0.40	39.8	46	+1/−1	700

Shaded = highest score. Δ columns compare with the pooled mean of unmodified S3-FIFO (1,241.4 ops/s). Class: real = counts toward the cap, tune = graded the same but not counted. Eval s = submit → graded. Status is CORAL's comparison with the agent's own previous best, not with a baseline.

Diagnostic probes and derived per-operation metrics

#	Commit	Policy	Bytes read MiB	Δ% vs S3-FIFO mean	Bytes written MiB	Read I/Os	Refaults	Δ% vs S3-FIFO mean	pgmajfault	pgactivate	Resident file MiB	Ops in phase (≈ tput × 30)	Read KB / op	Refaults / 1k ops	Read I/Os / 1k ops	Read MB/s
1	9d8172f7	get_scan.c (unmodified)	258.1	-12.4	0.0	23,416	39,762	-18.4	16	10,210	46	32,009	8.46	1,242.2	731.5	9.02
2	5f742d7a	s3fifo.c (unmodified)	294.5	+0.0	0.0	22,959	48,724	+0.0	31	13,957	47	36,500	8.46	1,334.9	629.0	10.29
3	463657d3	evolved	302.0	+2.5	0.0	21,690	50,612	+3.9	31	14,095	47	35,316	8.97	1,433.1	614.2	10.56
4	63f83a6a	evolved	289.2	-1.8	0.0	23,301	47,463	-2.6	31	13,570	46	37,036	8.19	1,281.5	629.1	10.11
5	f805b52d	evolved	303.6	+3.1	0.0	21,971	50,963	+4.6	31	14,429	48	36,301	8.77	1,403.9	605.3	10.61
6	2b1dc4db	evolved	308.6	+4.8	0.0	22,451	52,276	+7.3	31	14,517	47	37,203	8.70	1,405.2	603.5	10.79
7	545a9bfa	evolved	283.4	-3.8	0.0	23,113	45,999	-5.6	31	13,567	47	36,977	8.04	1,244.0	625.1	9.91
8	2403005d	evolved	286.4	-2.8	0.0	23,212	46,771	-4.0	31	13,006	47	36,977	8.12	1,264.9	627.7	10.01
9	3233ed4a	mglru.c (unmodified)	306.4	+4.0	0.0	22,461	51,767	+6.2	31	13,864	47	37,869	8.48	1,367.0	593.1	10.71
11	61e5e18b	evolved	293.9	-0.2	0.0	23,373	48,610	-0.2	16	13,506	46	36,994	8.33	1,314.0	631.8	10.27
12	19e7b6b1	mglru.c (unmodified)	293.5	-0.3	0.0	22,361	48,536	-0.4	31	13,354	47	36,383	8.46	1,334.0	614.6	10.26
13	44a8a8a9	fifo.c (unmodified)	293.9	-0.2	0.0	23,297	48,602	-0.3	16	13,677	45	36,819	8.37	1,320.0	632.7	10.27
14	11bac9cf	sampling.c (unmodified)	262.2	-11.0	0.0	22,518	40,639	-16.6	16	9,833	45	33,445	8.22	1,215.1	673.3	9.16
15	7efd90dc	lhd.c (unmodified)	263.8	-10.4	0.0	22,898	41,134	-15.6	16	10,359	46	33,093	8.36	1,243.0	691.9	9.22
16	d799dfdb	evolved	294.8	+0.1	0.0	23,695	48,821	+0.2	31	13,579	47	37,266	8.29	1,310.1	635.8	10.30
17	ca44f70f	evolved	283.9	-3.6	0.0	22,715	46,143	-5.3	31	13,113	47	36,158	8.23	1,276.2	628.2	9.92
18	9279fd33	evolved	299.6	+1.7	0.0	23,970	50,033	+2.7	31	13,764	47	37,032	8.48	1,351.1	647.3	10.47
19	d7fd5641	evolved	287.5	-2.4	0.0	22,880	46,978	-3.6	31	13,138	47	35,300	8.54	1,330.8	648.2	10.05

#	Commit	Policy	Bytes read MiB	Δ% vs S3-FIFO mean	Bytes written MiB	Read I/Os	Refaults	Δ% vs S3-FIFO mean	pgmajfault	pgactivate	Resident file MiB	Ops in phase (≈ tput × 30)	Read KB / op	Refaults / 1k ops	Read I/Os / 1k ops	Read MB/s
20	7f1ae77b	evolved	291.6	-1.0	0.0	23,520	48,015	-1.5	31	13,518	47	37,393	8.18	1,284.1	629.0	10.19

Reference = unmodified S3-FIFO (this run, attempt #2): bytes read 294.5 MiB, refaults 48,724. Assumes the probes cover exactly the 30 s measured phase, so operations = throughput × 30.

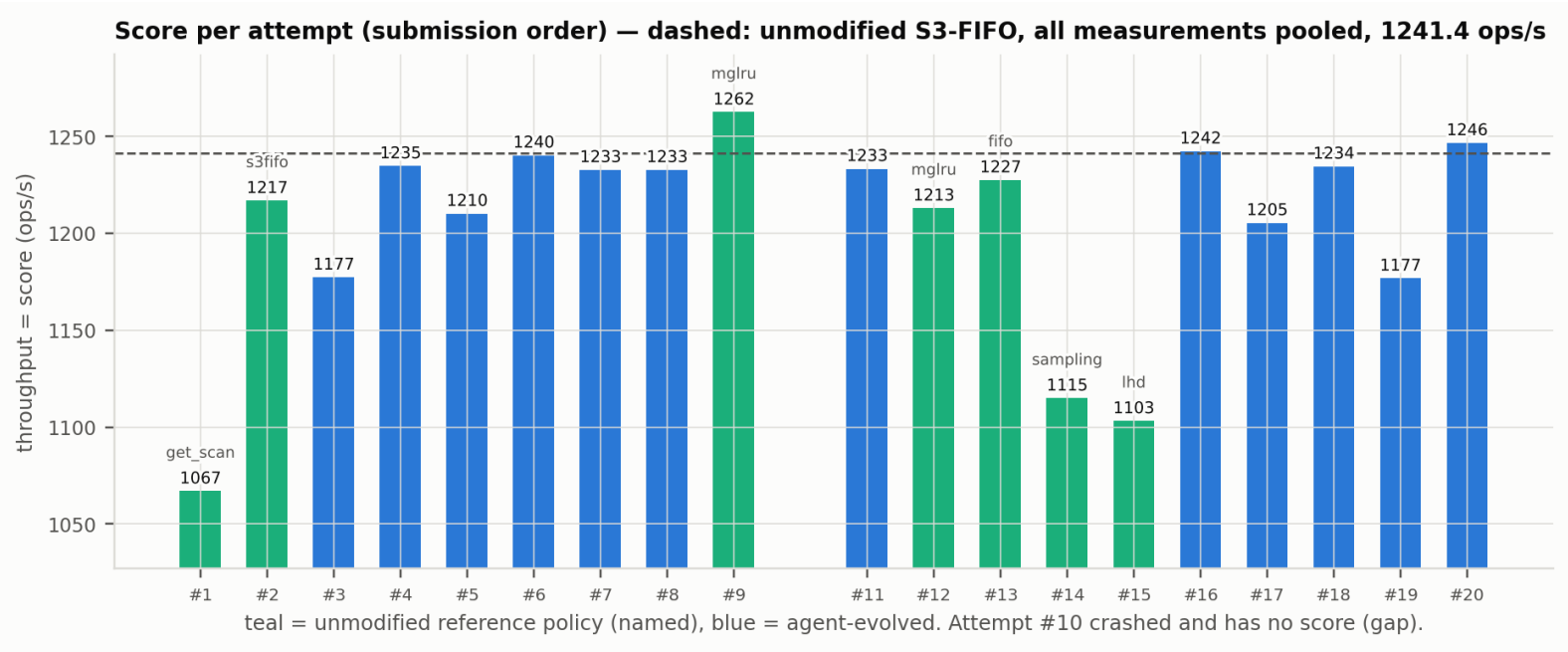
Attempts without a score (benchmark did not produce a result)

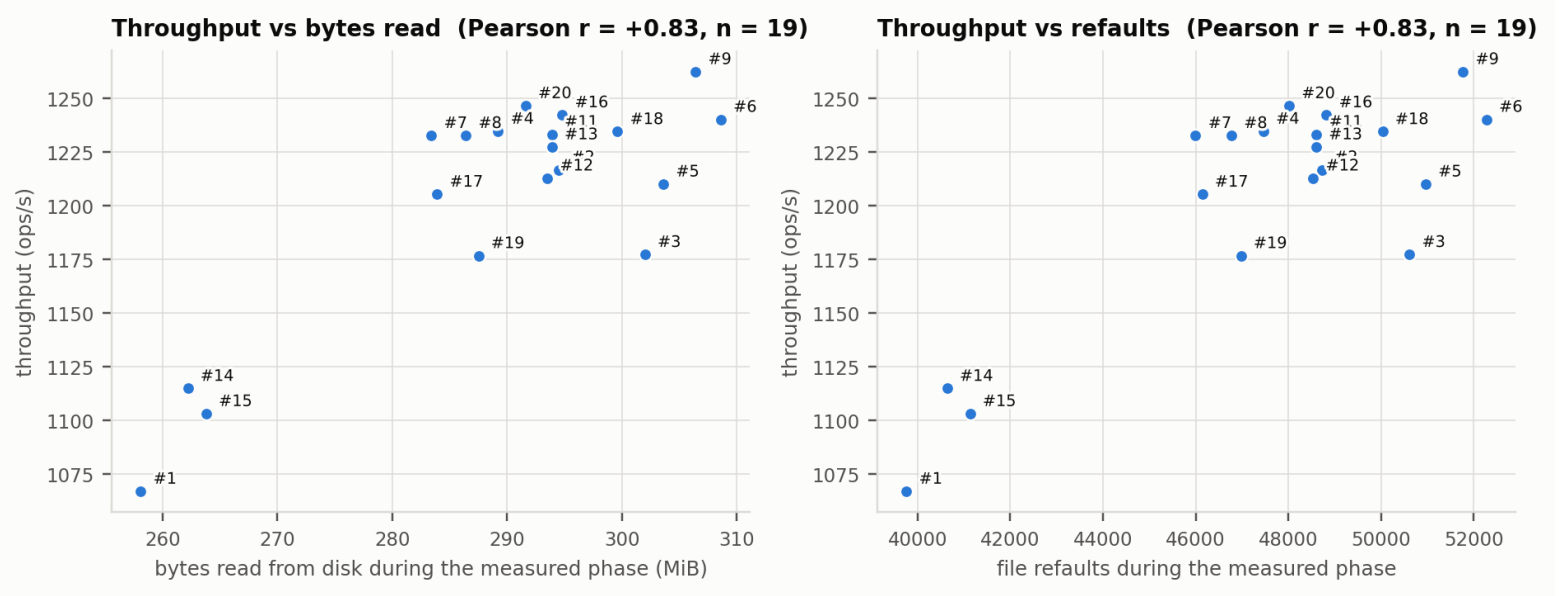
#	Commit	Class	Wall s	First line of grader feedback
10	dbea77c9	real	3.0	eval: [FAIL] wall=3.05s error=benchmark[all] returned 1

Attempt descriptions (commit messages, verbatim)

#	Commit	Class	Description
1	9d8172f7	real	baseline: reference get_scan.c unchanged (scan list + sampled LFU)
2	5f742d7a	real	baseline: reference s3fifo.c unchanged
3	463657d3	real	s3fifo + scan list: scan-pid inserts go to separate FIFO evicted first (>64 pages), scan-thread accesses don't bump freq, GET-reused scan pages promoted to main
4	63f83a6a	real	s3fifo+scan list: raise scan-list floor 64 -> cache_size/16 so readahead pages survive until scan consumes them
5	f805b52d	real	s3fifo+scan list: scan-list floor cache_size/16 -> /8 (probe direction of floor sensitivity)
6	2b1dc4db	real	confirm run: s3fifo+scan list with floor cache_size/16 (same code as 63f83a6a, which scored 1234.5)
7	545a9bfa	real	s3fifo+scan list(/16) + LevelDB index-block (last file page) admitted straight to main with freq=3
8	2403005d	real	hot-path trim: folio_accessed only does scan_pids lookup for scan-list folios (fewer hash lookups per GET); keeps index-block pin
9	3233ed4a	real	baseline: reference mglru.c unchanged (family comparison vs s3fifo ~1217)
10	dbea77c9	real (crashed)	mglru + scan list: scan-pid inserts go to separate FIFO evicted before generations when >768 pages (~cache/16, best floor from s3fifo sweep); scan-thread touches don't bump refs on scan pages
11	61e5e18b	real	mglru + scan list (retry): mask mglru_lists[] indices with &(MAX_NR_GENS-1) to fix verifier 'unbounded memory access' in evict_folios
12	19e7b6b1	real	confirm run: reference mglru.c unchanged (3233ed4a scored 1262.3, single run)
13	44a8a8a9	real	sensitivity probe: reference fifo.c unchanged (does policy choice matter at all? all policies so far 1210-1262 incl noise)
14	11bac9cf	real	family probe: reference sampling.c (sampled LFU, 20 samples, index-block bonus) unchanged; hypothesis: IRM/Zipf workload favors frequency over recency since fifo~s3fifo~mglru
15	7efd90dc	real	structural attempt 1/3 on LHD: reference lhd.c unchanged (only untested family; hit-density ranking with adaptive class stats)
16	d799dfdb	real	scan-favoring 1/3: invert priority - evict GET pages (s3fifo small/main) first, scan list only when GET pages < cache/4. Hypothesis: single scan thread bounds total ops (rbytes correlates + with throughput across 13 runs; LFU/LHD low rbytes -> low tput)
17	ca44f70f	real	scan-favoring 2/3: GET floor cache/4 -> cache/8 (give scan pages more cache; tests whether tput rises as scan share grows)
18	9279fd33	real	scan-favoring 3/3: confirm run of d799dfdb (GET floor cache/4, scored 1242.2) to separate signal from noise
19	d7fd5641	real	s3fifo small-FIFO share cache/15 -> cache/40: evict GET one-touch (cold Zipf tail) pages sooner, bigger main; base = d799dfdb (scan-favoring cache/4, mean 1238 over 2 runs)
20	7f1ae77b	real	new-page grace: s3fifo small-FIFO share cache/15 -> cache/6 (opposite of d7fd5641's /40 which lost 5%; 4 runs show evicting new pages fast hurts)

Score per attempt





Same code, several runs: is anything better than the references?

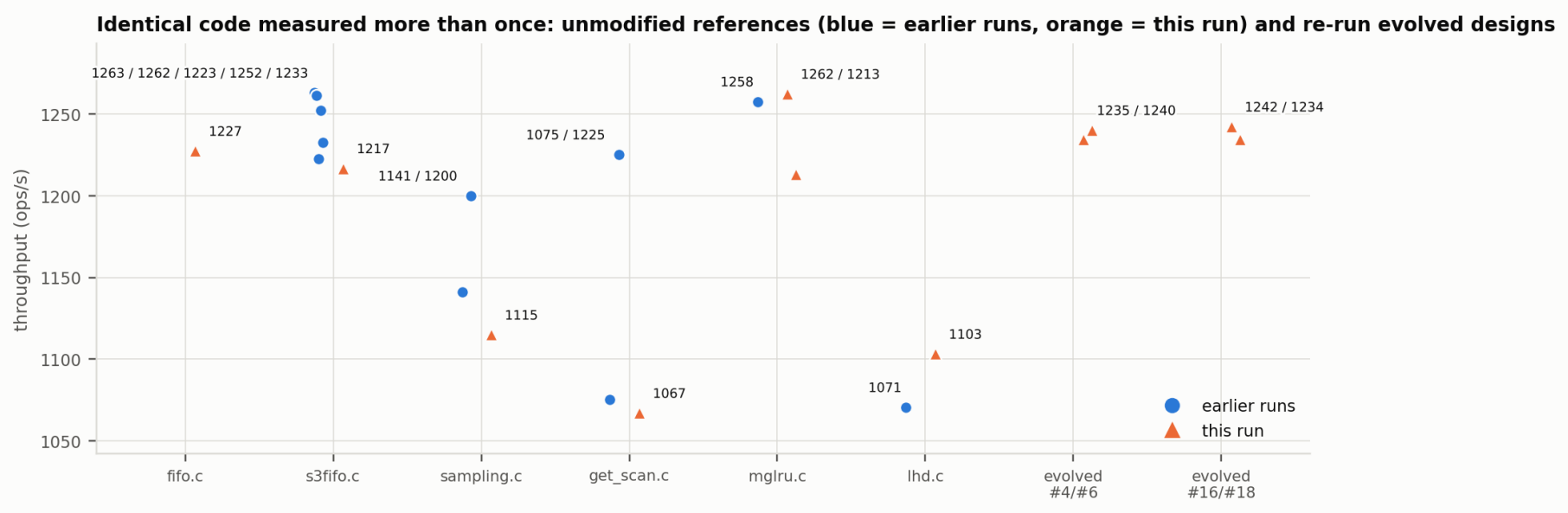
Unmodified reference code	Earlier runs (run: ops/s)	This run (#: ops/s)	Earlier mean	This run	Δ ops/s	Δ%
fifo.c	not run	#13: 1,227.31	-	1,227.3	-	-
mru.c	run 2: 1,258.7	not run	1,258.7	-	-	-
s3fifo.c	run 1: 1,263.0; run 1: 1,261.5; run 2: 1,222.5; run 3: 1,252.3; run 3: 1,232.5	#2: 1,216.66	1,246.4	1,216.7	-29.7	-2.4
sampling.c	run 1: 1,141.3; run 2: 1,200.0	#14: 1,114.84	1,170.6	1,114.8	-55.8	-4.8
get_scan.c	run 1: 1,075.1; run 2: 1,225.2	#1: 1,066.96	1,150.2	1,067.0	-83.2	-7.2
mglru.c	run 1: 1,257.6	#9: 1,262.31, #12: 1,212.78	1,257.6	1,237.5	-20.1	-1.6
lhd.c	run 1: 1,070.5	#15: 1,103.11	1,070.5	1,103.1	+32.6	+3.0

Runs: run 1 = z-score scoring (004013, 20 attempts); run 2 = absolute scoring, 5 attempts (102552); run 3 = single-seed S3-FIFO (162748). Only attempts whose code is byte-identical to the reference (comment-only lines ignored) are compared.

Identical code measured more than once within this run

Code	Measurements (ops/s)	n	Mean	sd	Range % of mean
Evolved, attempts #4/#6: s3fifo+scan list: raise scan-list floor 64 -> cache_size/16 so readahe	1,234.54, 1,240.09	2	1,237.3	3.9	0.45
mglru.c (unmodified)	1,262.31, 1,212.78	2	1,237.5	35.0	4.00
Evolved, attempts #16/#18: scan-favoring 1/3: invert priority - evict GET pages (s3fifo small/mai	1,242.21, 1,234.39	2	1,238.3	5.5	0.63

The unmodified S3-FIFO alone spans 1,216.7 … 1,263.0 ops/s (3.7% of its mean) over 6 measurements in four runs, so differences of a few percent between single attempts are within measurement noise.



Best attempt and best agent-evolved design

Item	Highest score (overall)	Best agent-evolved design
Attempt	#9 3233ed4a7506f3b792ae5f259490dc66118db99a	#20 7f1ae77bebedf6c0e9559b5232833c609daf3433
Policy	mglru.c (unmodified)	evolved (best agent-evolved attempt)
Score (= throughput, ops/s)	1,262.31	1,246.43
vs pooled unmodified S3-FIFO	+20.9 ops/s (+1.68%)	+5.0 ops/s (+0.40%)
Bytes read (MiB) / refaults	306.4 / 51,767	291.6 / 48,015
Code size	1109 lines (kernel 932, loader 177)	700 lines (kernel 477, loader 223)
Description	baseline: reference mglru.c unchanged (family comparison vs s3fifo ~1217)	new-page grace: s3fifo small-FIFO share cache/15 -> cache/6 (opposite of d7fd5641's /40 which lost 5%; 4 runs show evicting new pages fast hurts)

Ranking of all attempts by score

Rank	#	Commit	Policy	Score (ops/s)	Δ vs best (ops/s)	Δ vs best (%)
1	9	3233ed4a	mglru.c (unmodified)	1,262.31	+0.0	+0.00
2	20	7f1ae77b	evolved	1,246.43	-15.9	-1.26
3	16	d799dfdb	evolved	1,242.21	-20.1	-1.59
4	6	2b1dc4db	evolved	1,240.09	-22.2	-1.76
5	4	63f83a6a	evolved	1,234.54	-27.8	-2.20
6	18	9279fd33	evolved	1,234.39	-27.9	-2.21
7	11	61e5e18b	evolved	1,233.15	-29.2	-2.31
8	7	545a9bfa	evolved	1,232.56	-29.8	-2.36
9	8	2403005d	evolved	1,232.56	-29.8	-2.36
10	13	44a8a8a9	fifo.c (unmodified)	1,227.31	-35.0	-2.77
11	2	5f742d7a	s3fifo.c (unmodified)	1,216.66	-45.6	-3.62
12	12	19e7b6b1	mglru.c (unmodified)	1,212.78	-49.5	-3.92
13	5	f805b52d	evolved	1,210.02	-52.3	-4.14
14	17	ca44f70f	evolved	1,205.25	-57.1	-4.52
15	3	463657d3	evolved	1,177.20	-85.1	-6.74
16	19	d7fd5641	evolved	1,176.66	-85.6	-6.79
17	14	11bac9cf	sampling.c (unmodified)	1,114.84	-147.5	-11.68
18	15	7efd90dc	lhd.c (unmodified)	1,103.11	-159.2	-12.61
19	1	9d8172f7	get_scan.c (unmodified)	1,066.96	-195.3	-15.48

Each attempt is a single measurement; the unmodified S3-FIFO alone spanned 3.7% across 6 measurements, so adjacent ranks (differences under about 1–2%) are not distinguishable. The agent-evolved design's source is saved as best_evolved_policy.c.

Timing

Event	Time	Elapsed since start (min)
Run directory created / manager started	13:03:34	0.0
First model request at the gateway	13:03:40	0.1
First submission (attempt #1)	13:04:35	1.0
Last submission (attempt #20)	13:31:43	28.2
Last attempt graded / auto-stop recorded	13:32:34	29.0

Time budget	Seconds	% of run wall time
Evaluation (all 20 attempts, submit → graded)	874	50.2
Everything else (agent reading, thinking, editing)	866	49.8
Run wall time	**1,740**	**100.0**

Eval latency per attempt: min 44.6 s, median 46.0 s, max 46.6 s.

Agent sessions and LLM usage

#	Start	Wall s	API s (Δ)	Turns	Cost $ (Δ)	Input tok	Cache write tok	Cache read tok	Output tok	Thinking tok	API retries	Denials	Terminal reason
0	13:03:39	105	65	13	0.535	20	76,819	597,929	1,563	371	0	1	aborted_tools
1	13:05:24	120	34	6	0.146	10	8,036	321,688	2,049	234	0	2	aborted_streaming
2	13:07:25	110	57	5	0.165	8	8,940	266,295	3,339	690	0	0	aborted_streaming
3	13:09:15	75	22	4	0.085	6	4,762	192,887	1,115	546	0	0	aborted_streaming
4	13:10:30	70	16	3	0.040	4	2,105	99,212	463	95	0	0	aborted_streaming
5	13:11:40	60	7	3	0.039	4	2,128	101,317	425	68	0	0	aborted_streaming
6	13:12:41	75	16	3	0.055	4	2,107	103,445	1,187	450	0	0	aborted_streaming
7	13:13:56	85	35	3	0.052	4	2,851	105,552	826	375	0	0	aborted_tools
8	13:15:21	23	21	2	0.083	6	3,039	219,262	1,182	297	0	0	completed
9	13:15:46	75	8	4	0.064	6	2,561	224,690	327	47	0	0	aborted_streaming
10	13:17:01	65	51	7	0.255	12	13,174	596,890	3,508	490	0	1	aborted_tools
11	13:18:06	80	18	4	0.097	6	5,148	258,804	967	393	0	0	aborted_streaming
12	13:19:27	60	7	3	0.045	4	1,927	132,632	458	111	0	0	aborted_streaming
13	13:20:27	75	23	3	0.067	4	2,127	134,559	1,451	1,016	0	0	aborted_streaming
14	13:21:42	66	65	1	0.067	4	3,116	136,686	1,199	359	0	0	completed
15	13:22:52	105	55	8	0.250	14	7,818	855,695	1,962	1,008	0	0	aborted_tools
16	13:24:37	120	65	3	0.097	4	4,476	147,620	2,271	1,769	0	0	aborted_streaming
17	13:26:37	85	32	3	0.108	4	5,832	152,096	2,443	1,167	0	0	aborted_streaming
18	13:28:03	70	18	3	0.073	4	6,148	157,928	546	115	0	0	aborted_streaming
19	13:29:13	60	9	3	0.063	4	4,134	164,076	485	98	0	0	aborted_streaming
20	13:30:13	75	21	3	0.081	4	4,041	168,210	1,346	816	0	0	aborted_streaming
21	13:31:29	65	10	3	0.072	4	4,977	172,251	638	186	0	0	aborted_streaming
Σ		**1,719**	**656**	**90**	**2.54**	**140**	**176,266**	**5,309,724**	**29,750**	**10,701**	**0**	**4**	

Cost and API time are cumulative in Claude Code's result records, so rows show the increase over the previous session and Σ the final value; cost is Claude Code's estimate, not an AWS bill. Wall s, turns and tokens are per session. aborted_streaming = the session was cut off mid-response (consistent with CORAL interrupting and resuming the agent).

Agent actions

Tool	Calls
Bash	47
Write	1

coral command	Count
eval	24
log	7
diff	1
wait	1
show	1

Permission denials (actions the permission layer refused)

Session	Tool	Target (truncated)
0	Bash	{"command": "ls -R .claude/eval_logs/9d8172f77db0c7178712824b915cedac745bd7c7/ | head -30; cp reference/s3fifo.c noop.c && coral eval -m \"baseline: reference s
1	Bash	{"command": "mkdir -p .claude/notes/experiments; python .claude/skills/create-notes/scripts/stamp.py experiment --out .claude/notes/experiments/eval-1-get-scan-
1	Write	{"file_path": "/home/aanil/Desktop/ldos/arjua-forks/coral-cache-evolution/results/cache-ext-get-scan/2026-10-11_130334/agents/captain-nemo/.claude/notes/experim
10	Bash	{"command": "coral log --all --recent -n 1; h=$(coral log --all --recent -n 1 | grep -o '| [0-9a-f]\\{8\\} |' | tr -d '| '); coral show $h 2>&1 | head -60"}

LLM gateway (LiteLLM → Bedrock)

Requests by model and HTTP status

Model alias	200	400	Total	Success %
claude-opus-5-5	95	22	117	81.2
claude-sonnet-5	0	21	21	0.0
all	**95**	**43**	**138**	**68.8**

Failure causes

HTTP	Model alias	Count	Message (summarised from the gateway's error body)
400	claude-opus-5-5	22	messages.N.output_config: extra inputs not permitted
400	claude-sonnet-5	21	thinking.type.disabled not supported for this model

Response time of successful (200) requests

n	Mean s	p50 s	p90 s	p99 s	Max s	Sum s
95	8.81	4.42	19.45	64.45	64.72	837

Context the agent received

Instruction file injected into the agent's prompts	Requests containing it	of total
~/arjua-forks/CLAUDE.md	138	138
~/arjua-forks/…/2026-10-11_130334/agents/captain-nemo/CLAUDE.md	138	138

Gateway: 138 requests in 21 gateway sessions. The first file is the arjua-forks repository's own project instructions (the earlier throughput-profiling project), loaded because the agent's worktree sits below that directory; the second is the task's CLAUDE.md written by CORAL.

Measurements not available for this run

Quantity	Status
Noop baseline in this run	None evaluated; the comparison point here is unmodified S3-FIFO (this run plus earlier runs, pooled).
Per-operation latency (READ p99, SCAN p99), per-thread throughput	Written to a per-eval scratch directory that is deleted after each eval; not retained
Policy counters	Not exported by any attempt in this run
Agent notes	0 saved; the agent's attempts to write experiment notes were denied by the permission layer (see permission denials)
CPU utilisation / cycles during evals	Not collected